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Recent advances in magnetic reconnection in Mars' space environment

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Abstract

Magnetic reconnection has recently emerged as an important process in the Martian space environment. In contrast to Earth, Mars lacks a global intrinsic dipole field and possesses an induced magnetosphere shaped by solar-wind interaction with the ionosphere, localized crustal magnetic anomalies, and abundant planetary heavy ions. Recent measurements from Mars Global Surveyor, Mars Express, MAVEN, and Tianwen-1 have revealed reconnection signatures in several distinct regions, including the dayside induced magnetosphere, the induced magnetotail, the crustal-field mini-magnetospheres, and the upper ionosphere. These reconnection signatures include Hall-like magnetic perturbations, multi-species ion outflows, magnetic flux ropes, reconnection fronts, and electron pitch-angle changes indicative of magnetic topology reconfiguration. This review synthesizes recent observational and theoretical progress on Martian magnetic reconnection and discusses its implications for magnetospheric structure, plasma energization, auroral activity, and atmospheric ion escape. Particular attention is given to the effects of interplanetary magnetic field variability, heavy-ion composition, and ionospheric collisionality, which distinguish reconnection at Mars from more familiar terrestrial cases. Although existing observations show that reconnection is a recurrent and potentially important component of Mars–solar-wind coupling, its global occurrence rate, spatial extent, and long-term atmospheric impact remain poorly constrained. Future coordinated multi-spacecraft measurements and multi-scale simulations will be critical for linking local reconnection physics to global plasma transport and atmospheric evolution.

1 Introduction

1.1 Mars' space environment

Mars does not possess a global intrinsic dipole magnetic field [1], which makes the space environment of Mars differs fundamentally from that of Earth. Mars' space environment is controlled by the interaction between the solar wind, the interplanetary magnetic field (IMF), the Martian upper atmosphere and ionosphere, and localized crustal magnetic anomalies. This interaction produces an induced magnetosphere [2] and mini-magnetosphere in regions of strong crustal magnetic fields [3] (Fig. 1). The induced magnetosphere is highly variable [4] and strongly dependent on upstream solar wind [5] and

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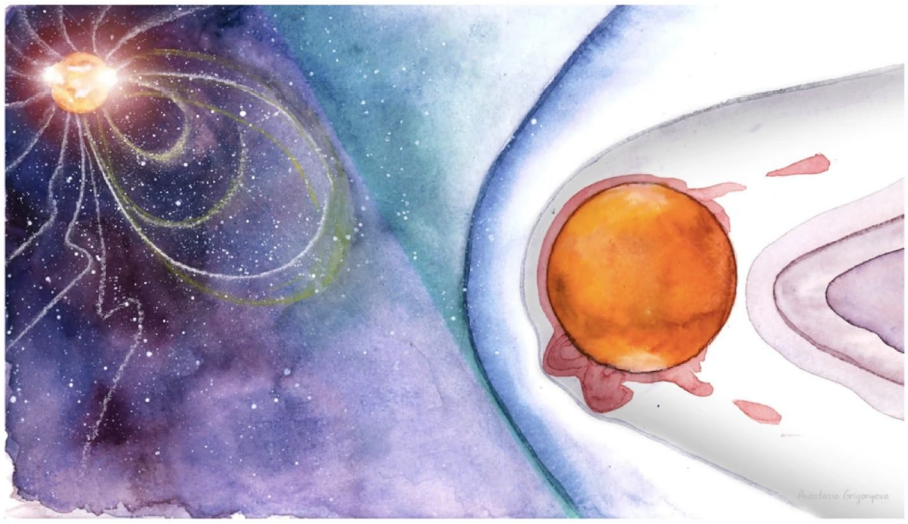


Fig. 1 Artistic diagram of the solar wind-Mars interaction adapted from Barabash, Holmström [10] (Grigoryeva, Anastasia. Solar Wind - Mars Interaction. Watercolour, 2015)

IMF [6, 7]. The mini-magnetosphere is shaped by both these external conditions [8] and contributions from crustal magnetic fields [9].

The induced magnetosphere forms because the solar wind interacts directly with Mars' ionosphere. The supersonic solar wind from upstream of Mars first encounters the planetary conducting layer and the planetary ions from the ionosphere mass-load the solar wind and slow the flow. At the same time, the IMF is compressed and draped around the ionospheric obstacle [11]. As the draped magnetic field is carried closer to Mars, it piles up above the ionosphere and forms the magnetic pileup region [1, 12]. The outer boundary of the induced magnetosphere is often associated with the magnetic pileup boundary [12], where the plasma composition and magnetic field properties change from solar-wind-dominated magnetosheath plasma to planetary-ion-rich induced magnetospheric plasma [13]. Boundaries between different regions on dayside induced magnetosphere are shown in Fig. 2. This process also creates an extended induced magnetotail on the nightside. In the magnetotail, the draped magnetic field is stretched anti-sunward into two lobes of oppositely directed magnetic field separated by a cross-tail current sheet [14]. The global position of the magnetotail current sheet also varies systematically with the radial IMF orientation, demonstrating an additional upstream control on the large-scale tail configuration [15].

The Mars' ionosphere is both a boundary of the induced magnetosphere and the main source of planetary ions. On the dayside, it is produced primarily by photoionization of the neutral upper atmosphere by solar EUV radiation [17]. On the nightside, where direct solar illumination is absent, ionization is maintained largely by plasma transport [18] and energetic electron precipitation [19, 20]. The ionosphere is dominated by heavy molecular and atomic ions, especially CO_2^+, O_2^+, and O^+ ions. And these ions can be supplied to the induced magnetosphere and magnetotail.

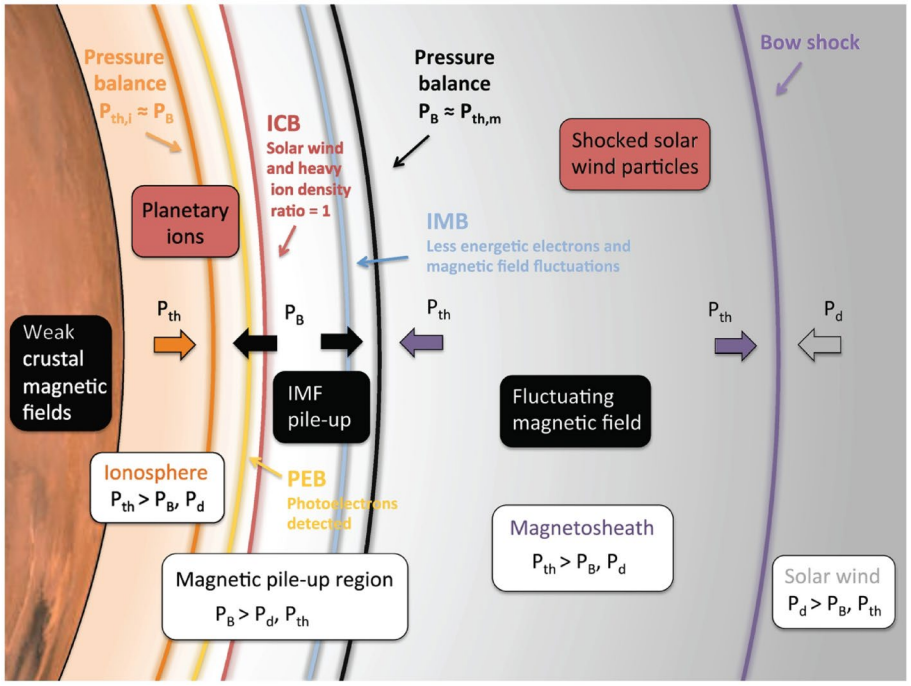


Fig. 2 Illustration of boundaries and regions on the dayside Mars' induced magnetosphere. This figure is adapted from Holmberg, André [16]

The presence of strong remanent crustal magnetic fields, mainly concentrated in the southern highlands (Fig. 3), is one of the distinctive features at Mars. These crustal magnetic anomalies form localized closed-loop systems or open magnetic field lines that can extend into the surrounding plasma environment (Fig. 4) [23]. They can modify the draped IMF [24–26], shield or trap ionospheric plasmas [27, 28], and guide electron precipitation [29]. As a result, the crustal magnetic fields provide additional magnetic boundaries and complex magnetic topologies around Mars. Recent statistical mapping further shows that the global distribution of magnetic field topologies in the Martian induced magnetosphere is systematically organized by the solar-wind convective electric field [30].

1.2 Magnetic reconnection physics and ion dynamics

Magnetic reconnection is a process in which magnetic field lines change their connectivity in a localized non-ideal region, enabling magnetic energy to be converted into plasma bulk motion, heating, and particle acceleration [32–35] (Fig. 5). In ideal magnetohydrodynamics (MHD), magnetic field lines are frozen into the plasma and magnetic topology is conserved. Reconnection occurs when this frozen-in condition breaks down, typically within thin current sheets where kinetic, resistive, or collisional effects become important [36]. The newly reconnected magnetic field lines are then accelerated away from the reconnection site by magnetic tension, producing bidirectional plasma outflows with speeds of the order of the local Alfvén speed.

In collisionless reconnection, the breakdown of the frozen-in condition occurs through the decoupling of ions and electrons. Because ions have larger inertial lengths

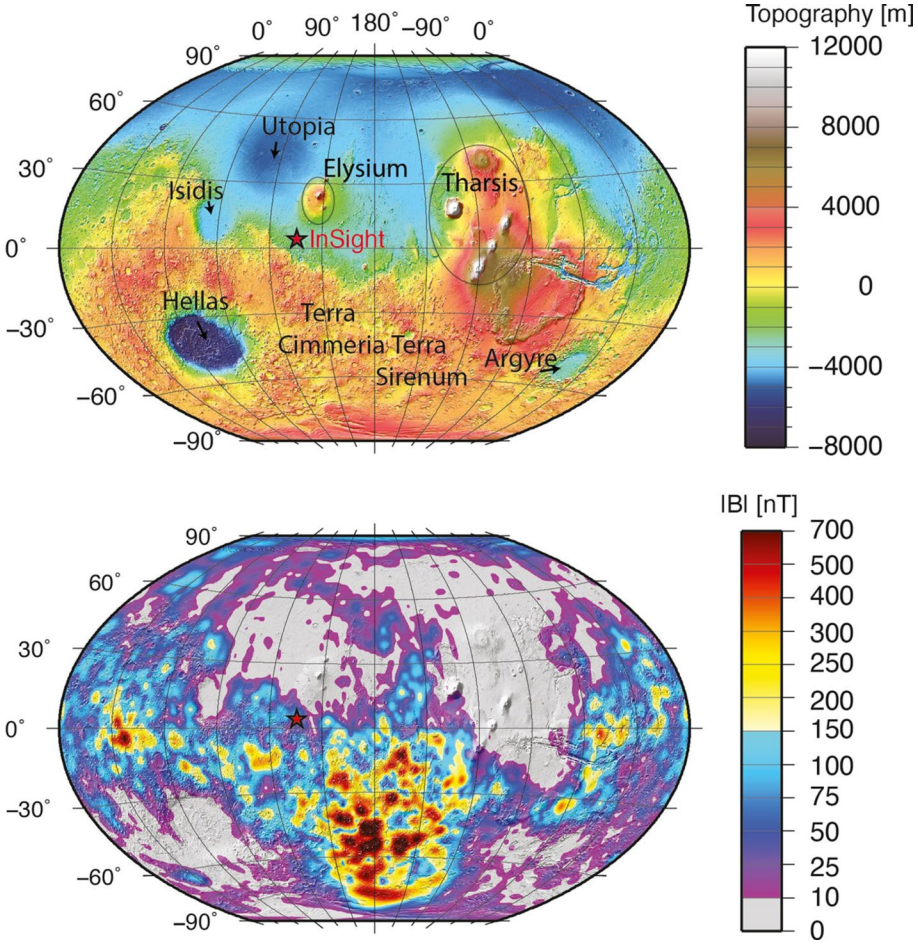


Fig. 3 The major physiographic features on Mars' surface and the crustal magnetic anomalies [1] predicted at 130 km altitude from Langlais, Thébault [21]. This figure is adapted from Mittelholz and Johnson [22]

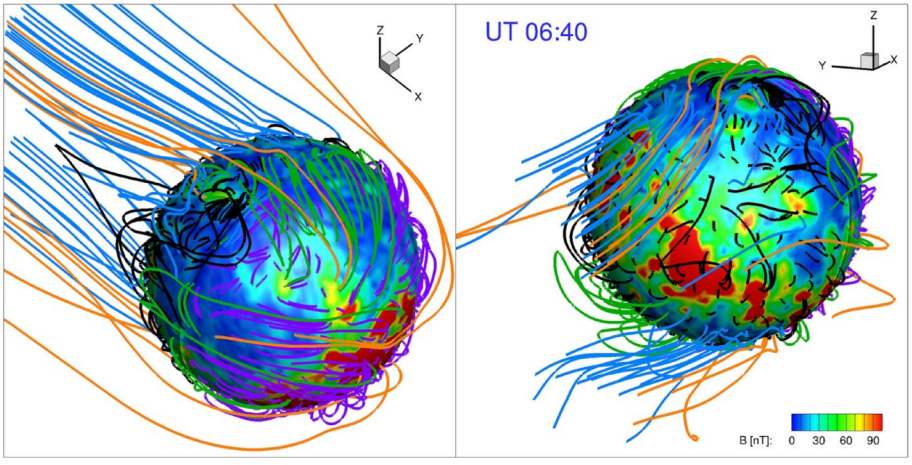


Fig. 4 Distribution of magnetic field lines around Mars extracted from the simulation result in Ma, Fang [31]. This figure is adapted from Xu, Mitchell [23]

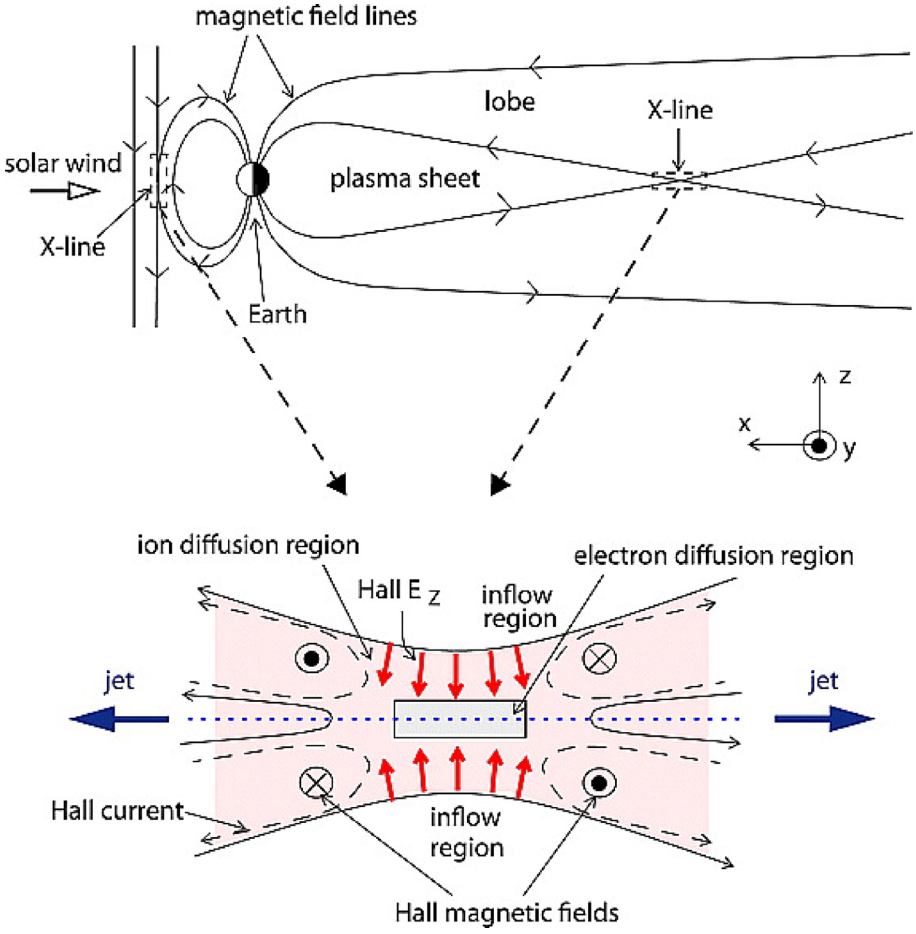


Fig. 5 Schematic diagram of magnetic reconnection in Earth's magnetosphere adapted from Paschmann [37]. Figure courtesy of Marit Oieroset

and gyroradii than electrons, especially CO_2^+, O_2^+, and O^+ ions, and these ions decouple from the magnetic field on ion scales, forming an ion diffusion region [38], while electrons remain magnetized until smaller electron scales are reached. This two-scale structure, consisting of an ion diffusion region and an embedded electron diffusion region [39–42] shown in Fig. 6, is a central feature of collisionless reconnection. Ion–electron decoupling also produces Hall currents and Hall magnetic perturbations, which are among the most widely used in-situ diagnostics of reconnection in planetary plasma environments [43–48].

Ion dynamics are central to the interpretation of reconnection observations. In the ion diffusion region and reconnection exhaust, ions can deviate from ideal $\boldsymbol{E}\times\boldsymbol{B}$ drift, become demagnetized, and gain energy from reconnection electric fields, Hall electric fields, and motional electric fields associated with reconnected magnetic flux. Reconnection outflows, ion beams, non-Maxwellian velocity distributions, and ion temperature enhancements are therefore important signatures of magnetic-to-particle energy conversion. In addition, secondary structures such as magnetic islands or plasmoids [50], flux ropes [51, 52], and reconnection fronts [53–56] can further energize ions through trapping, contraction, merging, and betatron- or Fermi-type acceleration [57, 58].

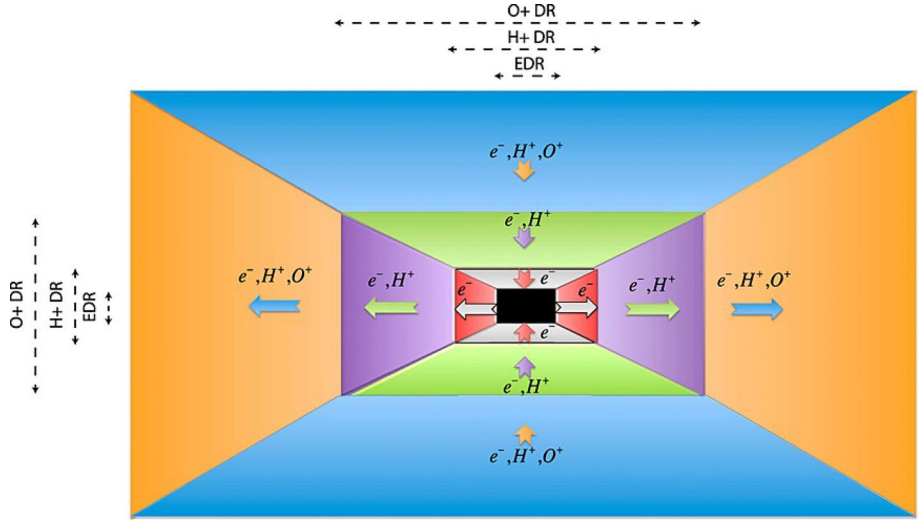


Fig. 6 Schematic of the multiscale reconnection diffusion region adapted from Liu, Mouikis [49]

For Mars, ion dynamics are expected to be important and different from Earth because the plasma is often composed not only of solar-wind protons but also of planetary heavy ions. Protons and heavy ions may not demagnetize or respond to reconnection electric fields in the same way, so reconnection at Mars often needs to be interpreted as a multi-ion process rather than as a simple electron–proton system. This makes ion composition a key parameter for understanding reconnection outflows, energy partition, and atmospheric ion escape at Mars. Besides, collisionality is another important factor. Reconnection in Earth's magnetosphere is generally expected to be collisionless. In Mars' lower ionosphere, however, ion-neutral and electron-neutral collisions become increasingly important and these collisions can modify the reconnection process. Therefore, Mars provides a natural environment in which reconnection may transition from collisionless behavior at higher altitudes to partially collisional behavior at lower ionospheric altitudes.

This review summarizes recent advances of magnetic reconnection in the Mars' space environment. We first outline the reconnection physics and Mars' plasma conditions most relevant to the observations. In the following sections, we review recent studies of reconnection in different regions of Mars, followed by a discussion of its effects on magnetospheric structure, ion escape, and heavy-ion reconnection physics. Finally, we summarize the current understanding of reconnection in Mars' space environment and propose several open questions for future studies.

2 Magnetic reconnection in Mars' space environment

2.1 Reconnection in the induced magnetosphere

The induced magnetopause (IMP), the outer boundary of the Mars' induced magnetosphere, separates shocked solar-wind plasma in the magnetosheath from the heavy-ion-rich plasma in the induced magnetosphere ([13, 59, 60]. Wang, Yu [61] presented in-situ evidence from MAVEN observations for magnetic reconnection at the dayside IMP as shown in Fig. 7. The event showed a clear transition from magnetosheath plasma to planetary-ion-dominated plasma across the boundary, together with magnetic signatures more consistent with a rotational discontinuity than with a purely tangential one.

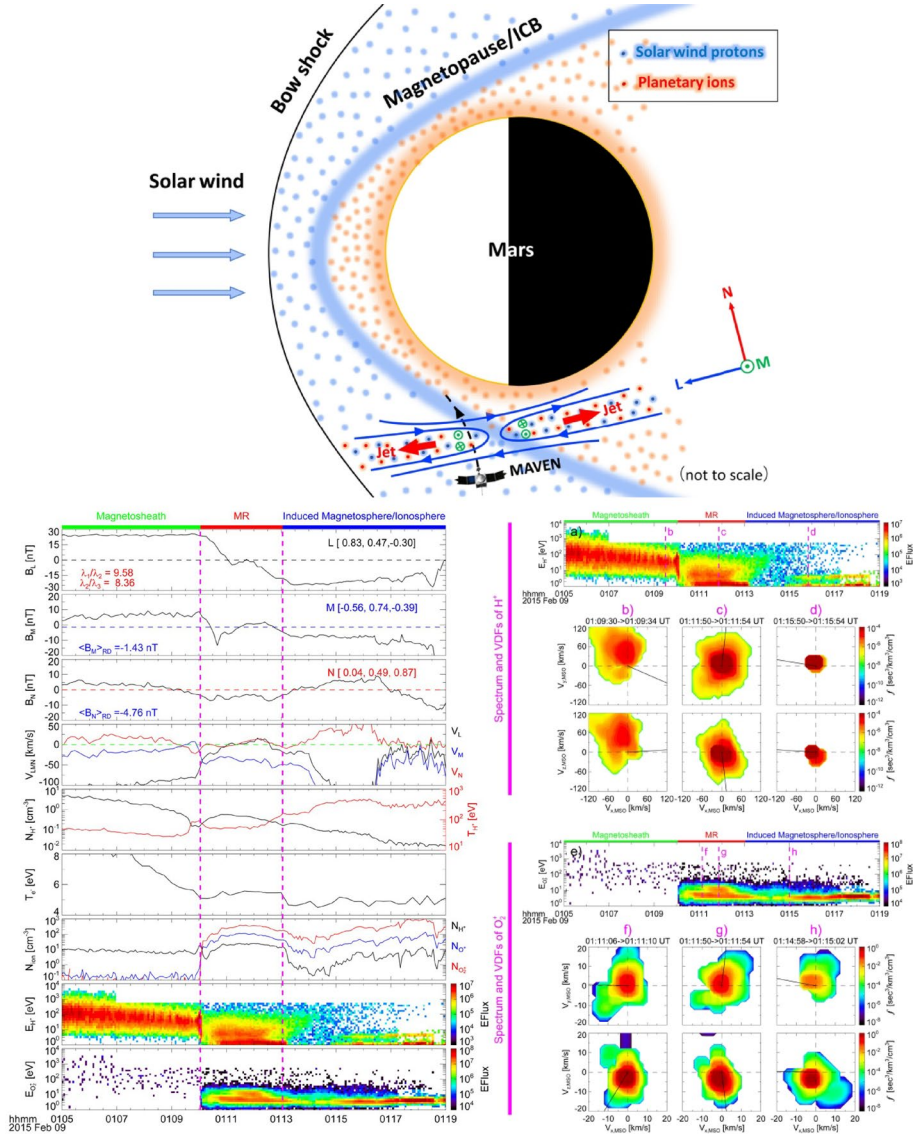


Fig. 7 Schematic illustration of magnetic reconnection at the Martian induced magnetopause/ion composition boundary (IMP/ICB) and observations from MAVEN during 01:05 – 01:19 UT on February 9, 2015. This figure is adapted from Wang, Yu [61]

Hall-like magnetic perturbations were also detected within the current sheet, and a sunward outflow jet was observed that contained both solar-wind protons and planetary heavy ions. These features suggest that the dayside induced magnetosphere can become magnetically open to the solar wind through reconnection.

This picture also implies that reconnection at the IMP is asymmetric, owing to the large contrast in plasma density and ion composition across the boundary (Fig. 7). On one side lies shocked proton-dominated solar-wind plasma, whereas on the other lies a denser planetary plasma dominated by heavy ions. Such asymmetry is expected to affect current-sheet structure, Hall-field geometry, and the relative acceleration of light and heavy ion species during reconnection [61, 62]. More broadly, IMP reconnection suggests that the dayside boundary cannot always be treated as an impermeable interface,

but may instead provide a direct pathway for plasma exchange and, potentially, for planetary ion loss through direct acceleration or subsequent solar-wind pickup.

Magnetic reconnection relevant to the near-Mars environment is not restricted to Martian plasma boundaries. MAVEN has recently provided direct evidence of Petschek-type reconnection exhausts within solar-wind current sheets near Mars, characterized by bifurcated magnetic-field profiles and Alfvenic ion [63]. The unusually broad exhaust suggests that reconnection can restructure and broaden upstream current sheets before they encounter the Martian bow shock. Such upstream preconditioning may modify the magnetic shear and plasma conditions incident on the induced magnetosphere, although its quantitative influence on IMP reconnection remains to be established.

2.2 Reconnection in the induced magnetotail

Similar to the magnetotail of Earth, Mars' induced magnetotail geometry provides favorable conditions for magnetic reconnection. Early evidence for magnetotail reconnection came from Mars Global Surveyor measurements, which revealed Hall-like magnetic signatures and electron pitch-angle features suggestive of reconnection diffusion regions [64, 65]. With the measurements from MAVEN, the observational picture became much clearer. Harada, Halekas [66] reported one of the first comprehensive event studies showing signatures consistent with collisionless reconnection in the near-Mars tail, including Hall magnetic perturbations, ion outflows involving both solar-wind protons and planetary heavy ions, and electron distributions indicative of altered magnetic topology (Fig. 8). Later work refined the observational criteria and showed that reconnection can be identified through a combination of current-sheet structure, Hall fields, multi-species ion jets, electron pitch-angle topology, and, where possible, Walén consistency of the outflow [67, 68]. These observations indicate that fully developed collisionless reconnection can occur in the Mars' magnetotail despite the absence of a global intrinsic dipole.

Statistical studies further suggest that magnetotail reconnection at Mars is not rare. Harada, Halekas [67] found that Hall-field signatures of reconnection occur preferentially in the near-Mars tail and show a pronounced hemispheric asymmetry, with more events detected in the – E hemisphere, in which the solar wind motional electric field points toward the planet (Fig. 9). A larger survey by [69], using both Hall signatures and fast proton flows, concluded that reconnecting current sheets make up a substantial fraction of Mars' magnetotail current-sheet crossings (Fig. 10). It showed that reconnecting sheets tend to be thinner than non-reconnecting ones, with characteristic thicknesses comparable to the local proton inertial scale, a condition favorable for fast collisionless reconnection [70, 71].

Coordinated MAVEN and Tianwen-1 observations further connect near-tail reconnection with downstream current-sheet dynamics. Wen, Halekas [72] found that approximately two-thirds of the magnetotail-flapping intervals observed by Tianwen-1 coincided with reconnection signatures detected by MAVEN, and several intervals contained multiple reconnection-generated flux ropes. They proposed that tailward-propagating flux ropes perturb the downstream plasma and drive instabilities that initiate current-sheet flapping. The association is statistically strong but should be described as

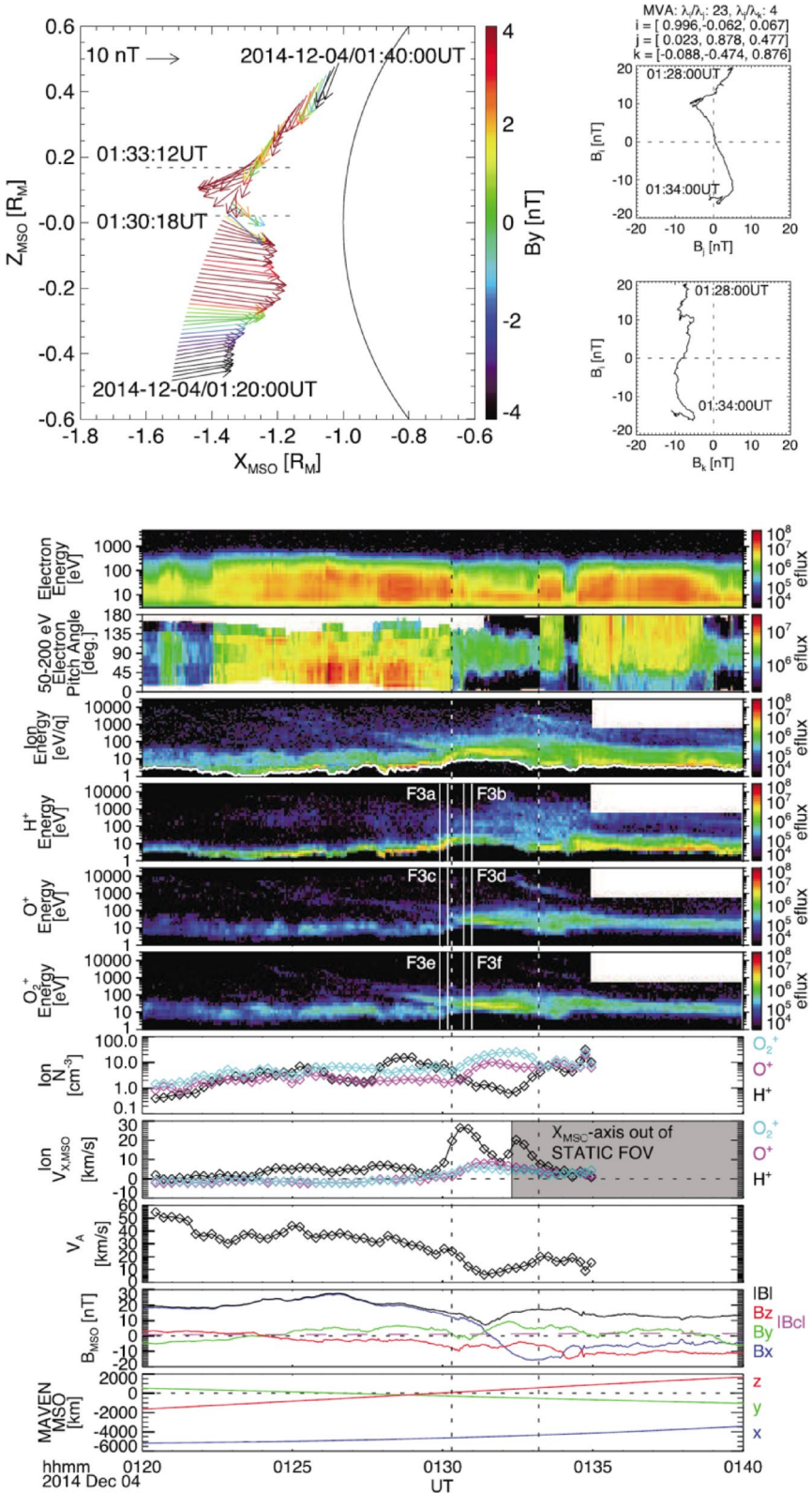


Fig. 8 Signatures of magnetic reconnection in the Mars' induced magnetotail from MAVEN measurements adapted from Harada, Halekas [66]

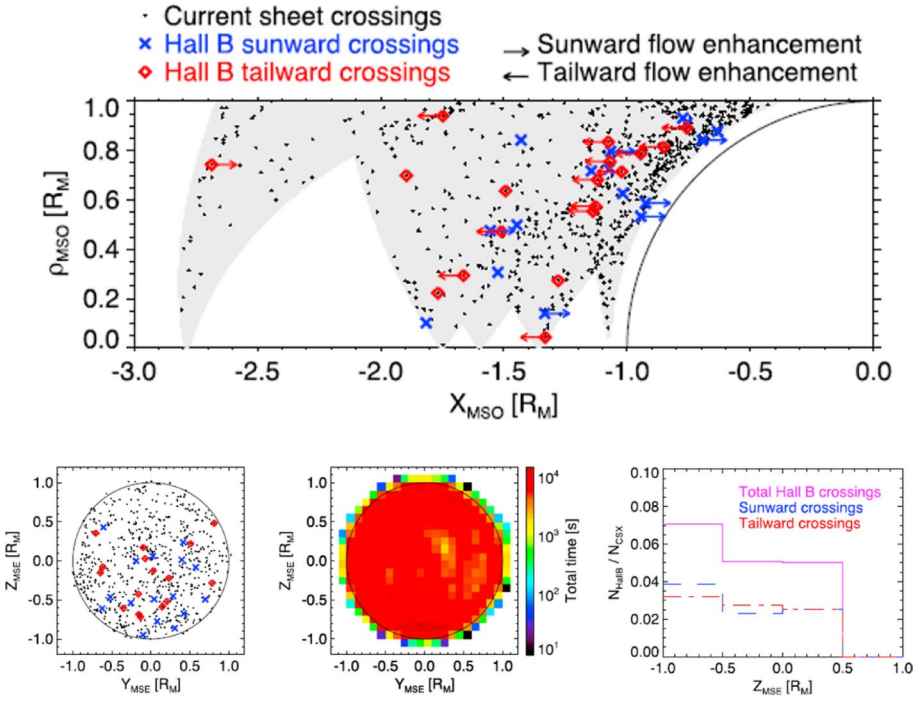


Fig. 9 Spatial distributions and observational frequency of current sheet crossings and Hall magnetic field events identified from MAVEN observations reproduced from Harada, Halekas [67]

a potential trigger rather than definitive causality, because a substantial fraction of flapping events occurred without a clear upstream reconnection signature.

A distinctive feature of Martian magnetotail reconnection is the strong presence of planetary heavy ions, especially O^+ and O_2^+, in addition to solar-wind protons. This multi-ion composition modifies the kinetic physics of reconnection relative to Earth's more proton-dominated plasma sheet. Observations consistently show that proton outflows are more strongly accelerated than heavy-ion outflows, reflecting the fact that lighter ions remain more effectively coupled to proton-scale reconnection structures, whereas heavier ions are weakly magnetized and respond differently to the reconnection electric field [49, 66, 67]. Statistical results likewise suggest that reconnecting current sheets are associated with enhanced tailward proton fluxes, whereas the response of heavy ions is less systematic [69]. This suggests that magnetotail reconnection likely contributes directly to proton loss and may also affect heavy-ion escape under specific conditions, for example through flux-rope formation, topology change, or indirect energization pathways [68, 73].

Planetward ion populations provide an additional, although non-unique, diagnostic of plasma transport in the Martian magnetotail. Global MAVEN flux maps reveal substantial inward as well as outward planetary-ion fluxes, while surveys of current-sheet crossings show that sunward or planetward ion jets are widespread and generally embedded in magnetic configurations consistent with reconnection [74, 75]. During the December 2022 disappearing-solar-wind event, high-energy heavy-ion fluxes decreased while low-energy fluxes increased; plume escape weakened but the net tailward escape flux increased [76], demonstrating that upstream dynamic pressure can strongly reorganize ion transport without establishing a one-to-one correspondence between planetward flow and reconnection.

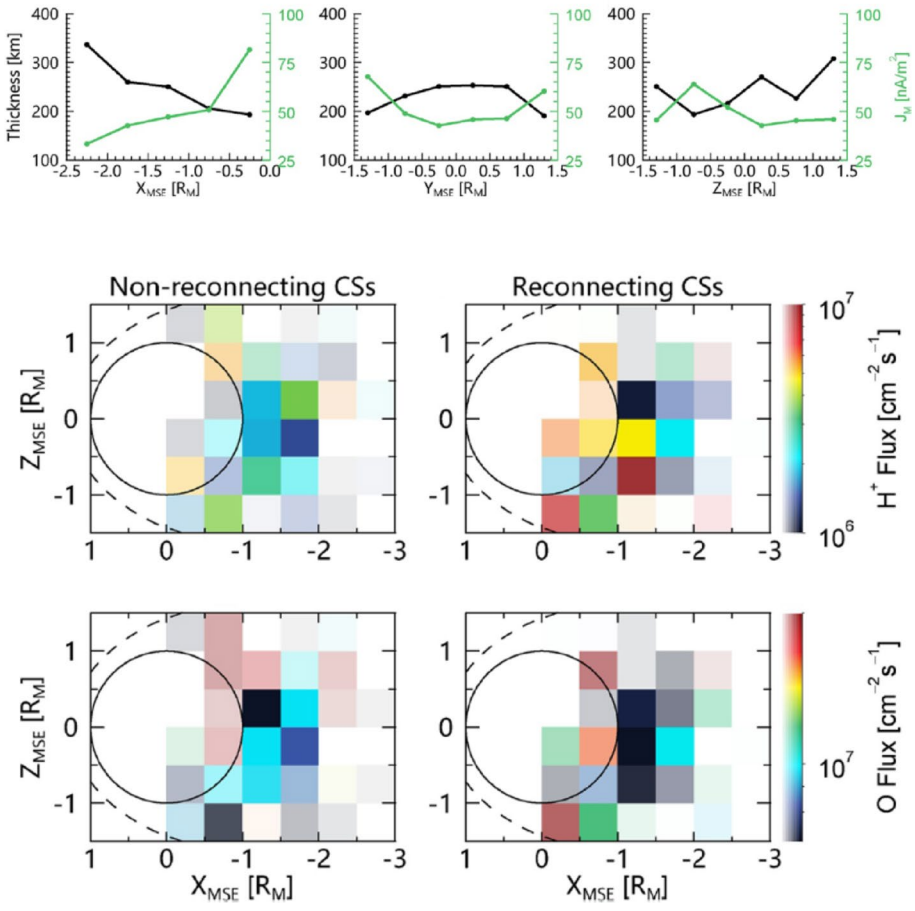
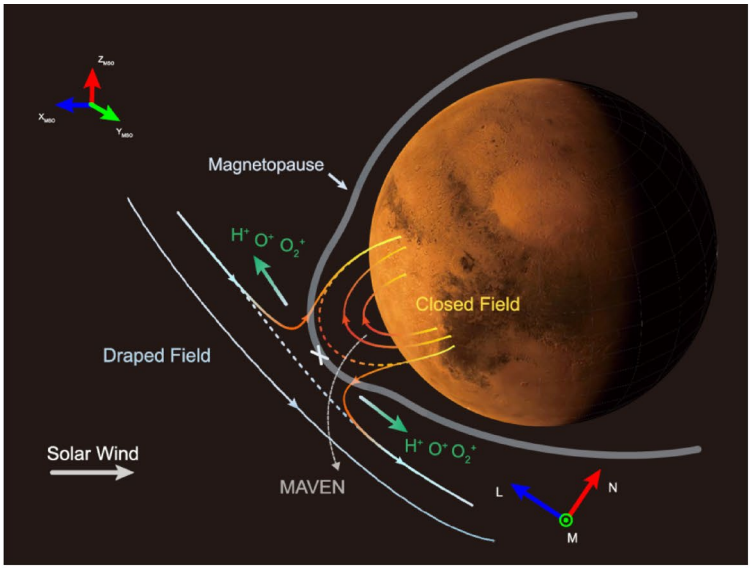


Fig. 10 The median of thickness (black curve) and current density J_M (green curve) of the current sheet as a function of the event locations and the mean values of ion net tailward fluxes in the MSE coordinates adapted from Wang, Huang [69]. The left (right) panels show the fluxes for non-reconnecting (reconnecting) CSs. The color opacity in each bin is determined by the event number within it and the normalized reference number is 20 (10) for non-reconnecting (reconnecting) CSs

2.3 Reconnection in crustal field regions

Mars' crustal anomalies interact directly with the draped IMF and the shocked solar wind [3, 77], which provides favorable conditions for magnetic reconnection.

Current observations indicated that reconnection in crustal-field regions occurs in several geometries. One is reconnection at the mini-magnetopause, where closed crustal magnetic fields interact with the antiparallel draped IMF [78, 79]. These events occur at the boundary between the shocked solar wind and crustal-field-supported mini-magnetospheres, and they share some features with boundary reconnection elsewhere in planetary plasma environments while remaining strongly influenced by local crustal topology and ionospheric plasma conditions (Fig. 11). A second geometry involves reconnection between neighboring crustal loops. Based on MAVEN observations, Chen, Huang [80] proposed that adjacent crustal loops can be sheared by solar-wind-driven flow, particularly near the terminator, where thin current layers may form and reconnect (Fig. 12). Lin, Huang [81] reported observations of interchange reconnection between open and



case from Lin et al. (2024, GRL) case from Qiu et al. (2024)

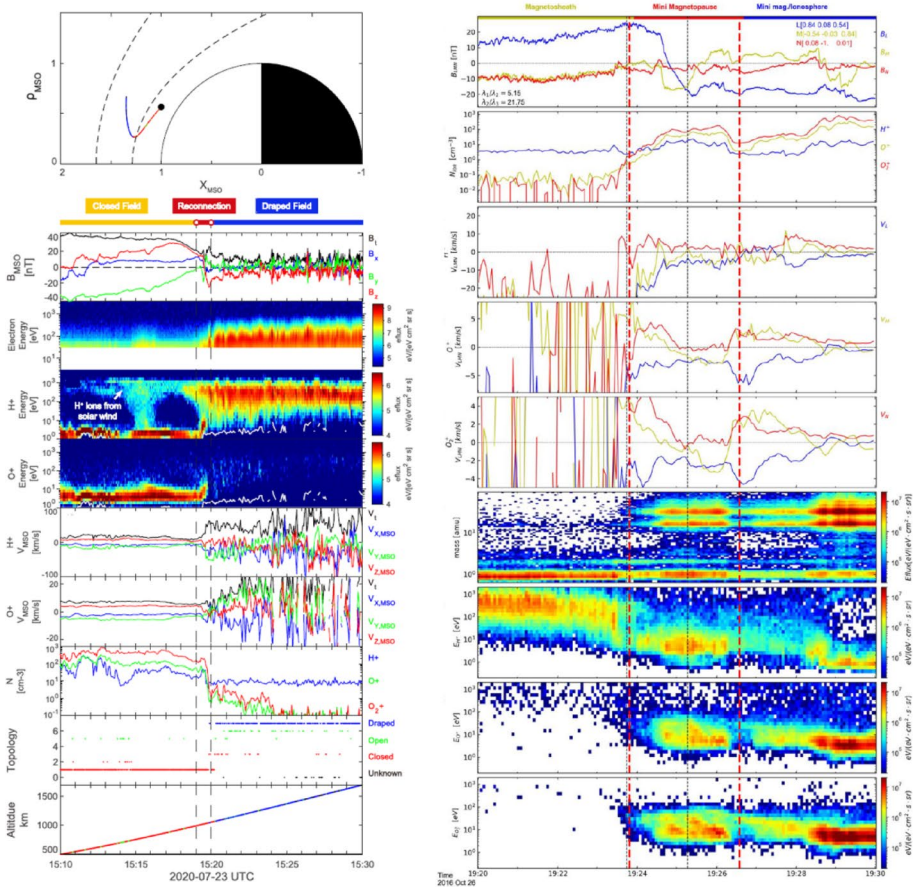


Fig. 11 Cartoon of magnetic reconnection between the crustal field and the draped field at mini-magnetopause and observational evidence of magnetic reconnection observed by MAVEN. This figure is revised from Lin, Huang [78] and Qiu, Yu [79]

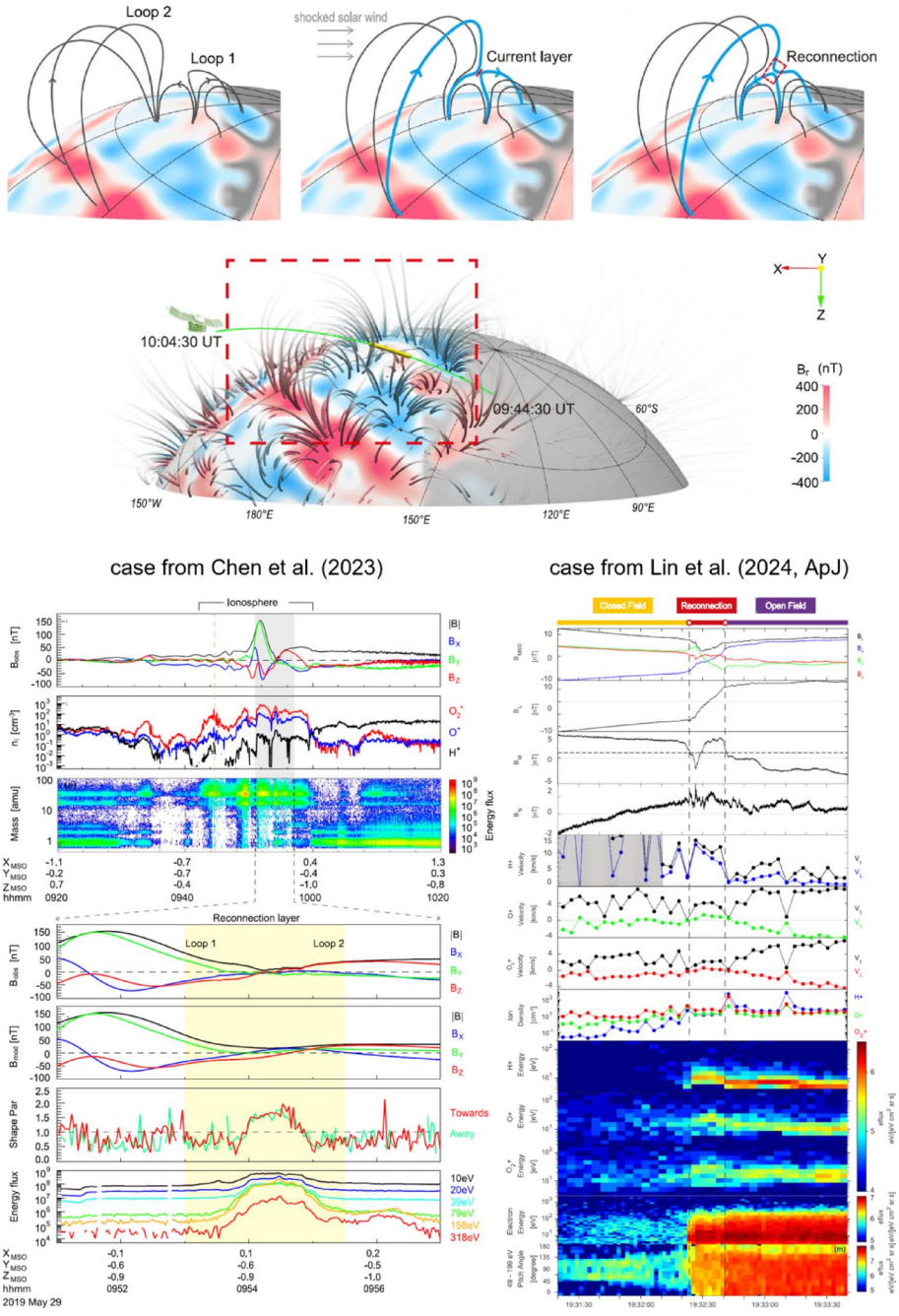


Fig. 12 Cartoon of interloop reconnection and observational evidence of interloop and interchange reconnection observed by MAVEN. This figure is revised from Chen, Huang [80] and Lin, Huang [81]

closed crustal fields in the near-Mars magnetotail (Fig. 12). A third category includes more eruptive or failed-eruptive events: Ye, Xu [82] interpreted a set of explosive ionospheric mass ejections as reconnection-driven events in which open crustal field lines reconnect and expel ionospheric plasma (Fig. 13), whereas Man, Xu [83] described dayside events with relatively weak outflows but substantial heavy-ion heating.

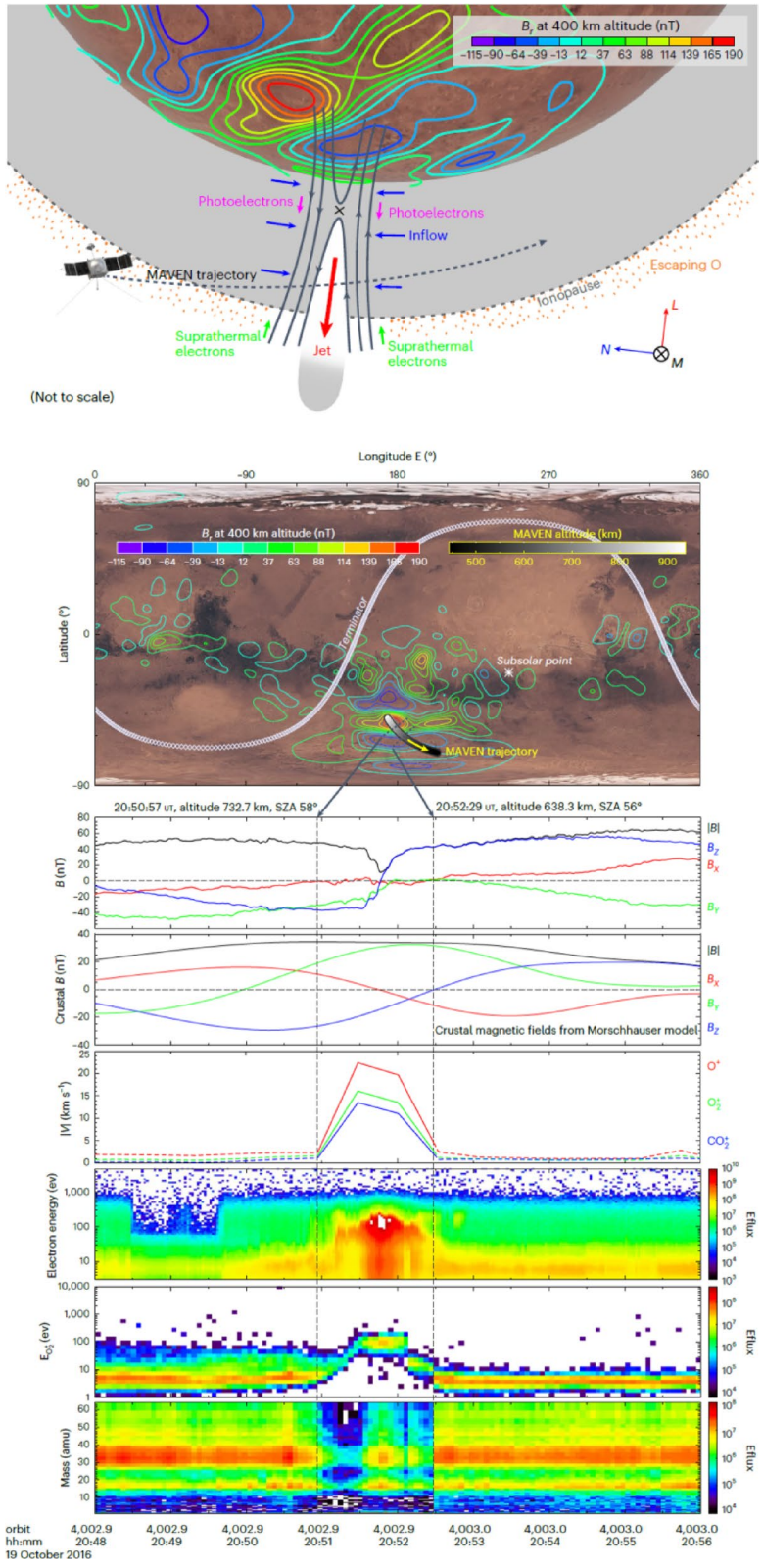


Fig. 13 Diagram of plasma cavity within a reconnection exhaust around strong crustal field regions adapted from Ye, Xu [82]

Reconnection in crustal-field regions has several important consequences for the Mars' plasma environment. First, it can open previously closed magnetic loops and thereby provide pathways for plasma exchange between the ionosphere and the surrounding solar-wind-driven environment [78–80]. Second, crustal-field reconnection is closely linked to discrete aurora. Statistical analysis of MAVEN observations shows that most discrete auroral events above the strongest southern crustal anomalies occur under field configurations favorable for antiparallel reconnection between crustal fields and the draped IMF (Fig. 14) [84], supporting the view that reconnection produces the open magnetic geometry and field-aligned electron acceleration needed for auroral precipitation [85–87]. Third, crustal-field reconnection may influence larger-scale magnetic connectivity by contributing to plasma transport into mini-magnetospheres and possibly to the broader magnetic configuration of the Martian tail and wake [24, 27, 88]. Although the quantitative importance of these effects remains uncertain, recent observations indicate that crustal-field reconnection is a significant component of Mars–solar-wind coupling.

2.4 Reconnection in the ionosphere

Mars' ionosphere serves both as the main source of planetary heavy ions and as a plasma regime that spans collisionless, weakly collisional, and strongly collisional conditions. Observational evidence indicates that magnetic reconnection can occur in the upper Mars' ionosphere in both strong and weak crustal-field settings. Above crustal-field regions, MAVEN observations reported by Ye, Xu [82] and Man, Xu [83] showed signatures consistent with collisionless reconnection, including thin current sheets, topology transitions inferred from suprathermal electrons, Hall-like magnetic perturbations, ion outflow jets, and localized ion energization. An important more recent development is the identification of reconnection in the dayside upper ionosphere above weak crustal-field regions. Using coordinated Tianwen-1 and MAVEN observations, Lin, Huang [89] reported a current-sheet crossing below the ion composition boundary that exhibited magnetic reversal, Hall-like magnetic structure, ion outflow approaching the local Alfvén speed, and a clear transition from closed field lines below the sheet to draped IMF-connected topology above it (Fig. 15). The event was linked to an upstream IMF rotation observed by Tianwen-1, suggesting that ionospheric reconnection is not only

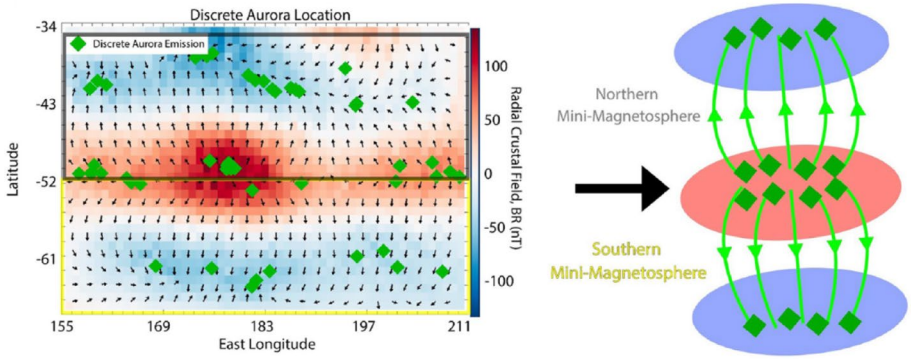


Fig. 14 Local map of discrete aurora emission events and the radial crustal field strength adapted from Bowers, Dibraccio [84]

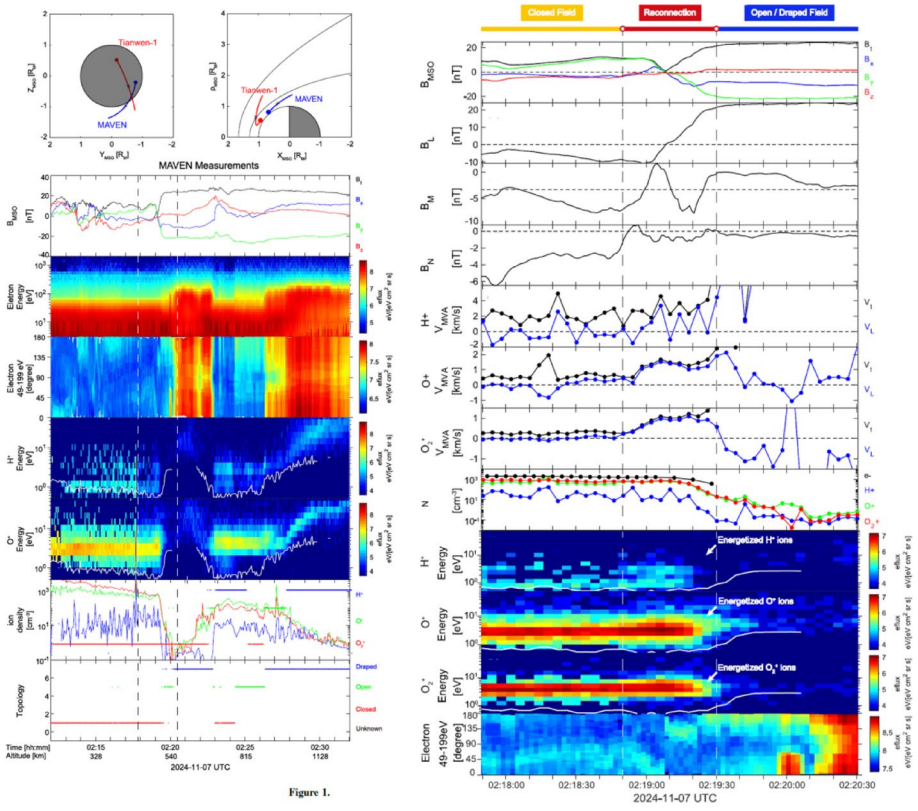


Fig. 15 Overview and evidence of ionospheric reconnection event above weak crustal field regions adapted from Lin, Huang [89]

confined to strongly magnetized crustal regions and may also be triggered more broadly by external magnetic forcing.

A distinctive aspect of reconnection in the Martian ionosphere is the role of collisions between charged particles and neutrals. As altitude decreases, increasing neutral density enhances ion-neutral and electron-neutral collisional coupling, which can suppress Hall physics, reduce outflow speed, and slow reconnection relative to the collisionless upper ionosphere [90]. This distinguishes the ionosphere from the weakly collisional plasma environments of the Martian magnetotail and outer induced magnetosphere. Cravens, Fowler [90] argued that the transition between collisionless and collisionally damped reconnection occurs near an altitude of about 300 km on the dayside, with efficient X-line reconnection favored above this level and increasingly suppressed reconnection below it. More recent statistical work by Harada, Cravens [91] supports this picture, showing that reliable ion-jet signatures are much more commonly identified in the collisionless regime than in the collisional one (Fig. 16). Even near the transition region, observed outflows tend to be sub-Alfvénic, consistent with partial damping by ion-neutral friction.

These ionospheric processes have important implications for planetary ion escape because they can convert cold, thermal ionospheric ions into energized, field-aligned outflows. This is particularly significant for heavy ions such as O^+ and $O_2{}^+$, which dominate much of the Martian ionosphere but require substantial energization to escape.

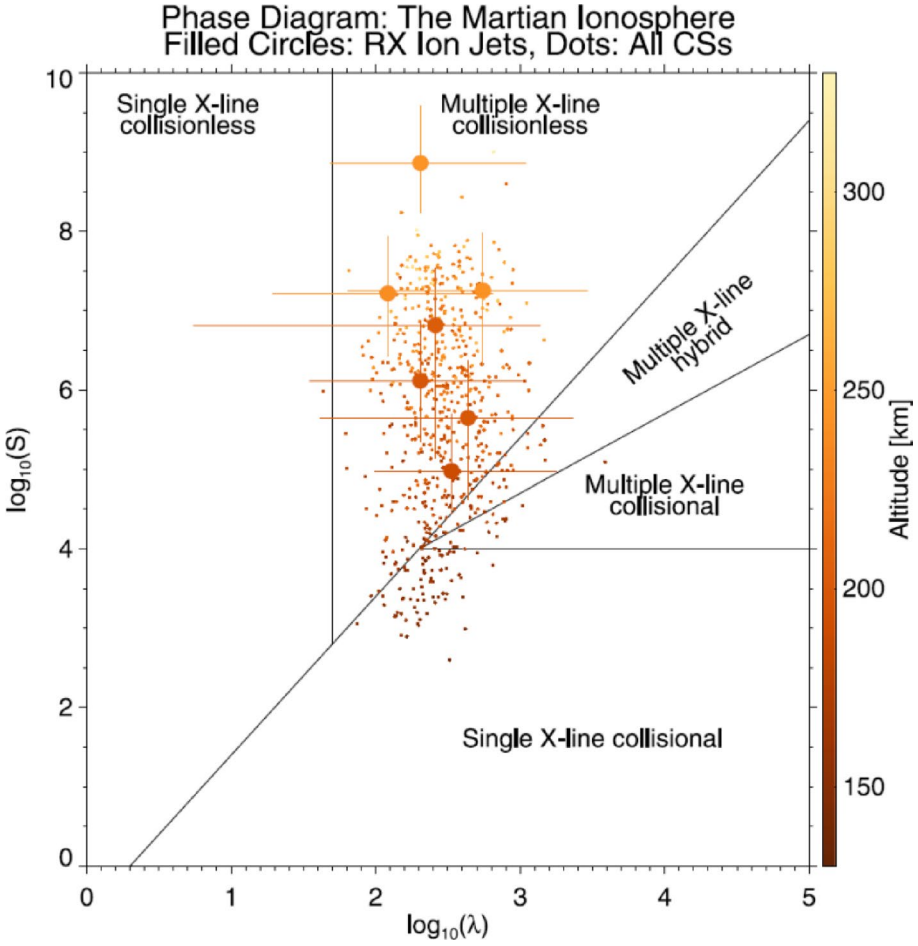


Fig. 16 Magnetic reconnection events (filled circles) and current sheets (small dots) observed by MAVEN in the Martian ionosphere compared to the phase diagram for magnetic reconnection from Ji and Daughton [92] and Ji and Daughton [93]. This figure is adapted from Harada, Cravens [91]

Recent event studies suggest that reconnection in the upper ionosphere can generate transient ion escape fluxes that are locally large compared with background outflow channels [89]. More broadly, ionospheric reconnection may act as a pre-acceleration process that feeds ions into larger-scale escape pathways, including transport into the magnetotail and pickup by the solar wind. Because this process can be externally driven by IMF changes, its long-term significance likely depends on solar-wind variability and may have been greater in the early Solar System, when both solar activity and atmospheric forcing were stronger.

2.5 IMF control of reconnection at Mars

Because Mars lacks a global intrinsic dipole field, the IMF provides the dominant external control on magnetic reconnection throughout the Martian plasma environment. As the IMF is compressed and draped around the planet, its orientation and temporal variability strongly influence where antiparallel field configurations develop, how current sheets are formed, and when reconnection is triggered. In this sense, reconnection at Mars is not controlled primarily by an internal global magnetic geometry, as at Earth, but by the coupling between upstream IMF conditions, induced magnetic topology, and localized crustal fields.

Evidence for this control has strengthened considerably in recent years. Early observations established that reconnection occurs in the Martian plasma environment, especially in the magnetotail [64], but more recent multipoint measurements have gone further by linking upstream IMF variability directly to reconnection onset. Using upstream observations from Tianwen-1 together with in-situ measurements from MAVEN, Wen, Halekas [68] reported a magnetotail reconnection event that followed a rapid IMF rotation by about two minutes, suggesting that upstream field rotations can reorganize the induced magnetotail and create favorable conditions for reconnection (Fig. 17). A similar coupling was reported in the dayside upper ionosphere, where Lin, Huang [89] observed reconnection signatures shortly after an IMF rotation measured upstream (Figs. 15 and 18). These studies both indicate that transient IMF variations can trigger reconnection in both the magnetotail and the dayside ionosphere.

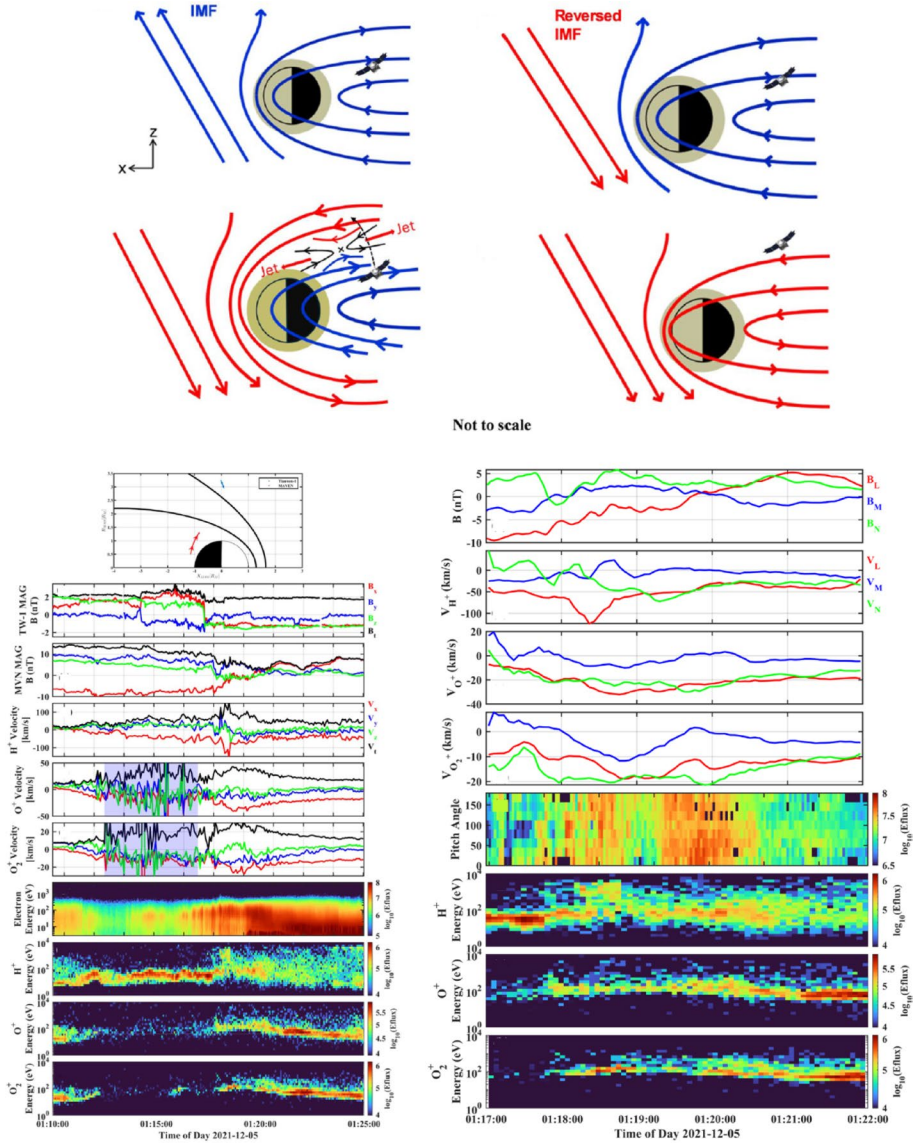


Fig. 17 Illustration of the reconfiguration of Mars' induced magnetosphere in response to the IMF rotation and simultaneous observations by Tianwen-1 and MAVEN. This figure is adapted from Wen, Halekas [68]

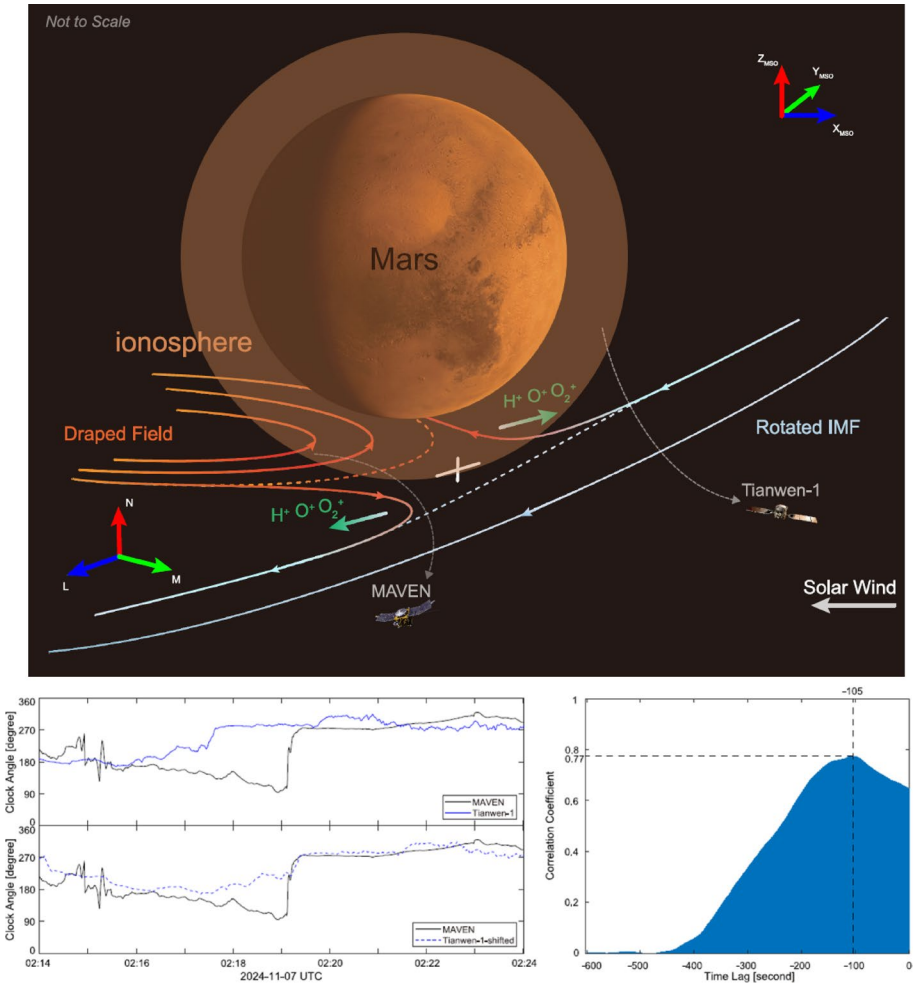


Fig. 18 Cartoon of magnetic reconnection in the ionosphere when IMF rotating and the cross-correlation analysis on the clock angle of magnetic field measured by Tianwen-1 and MAVEN. This figure is adapted from Lin, Huang [89]

The physical picture that emerges is that IMF rotations modify the draping pattern around Mars and can rapidly change the local magnetic shear at boundaries and within the induced magnetosphere (Fig. 18). This process can produce favorable antiparallel configurations, open magnetic field lines, and accelerate both solar-wind and planetary ions. In this way, IMF variability appears to link upstream solar-wind dynamics directly to ionospheric restructuring, magnetotail activity, and transient plasma escape from Mars. At the same time, the global significance of IMF-controlled reconnection remains uncertain. Most existing evidence still comes from a limited number of well-characterized events, and the relative importance of IMF forcing compared with local magnetic topology, crustal fields, and internal current-sheet evolution is not yet fully resolved. Nevertheless, current observations suggest that IMF variability is a major control on when and where reconnection occurs at Mars, and that its influence may have been even greater under the stronger and more variable solar-wind conditions of early Mars.

2.6 Numerical modeling perspectives on magnetic reconnection at Mars

Numerical studies provide an important complement to the in-situ observations reviewed above because they help link local reconnection signatures to the global solar wind–Mars interaction. Although direct numerical studies dedicated specifically to magnetic reconnection at Mars are still relatively limited, recent MHD, Hall-MHD, and hybrid investigations have significantly improved the physical context in which Martian reconnection should be understood [31, 94–96]. In particular, these studies show that reconnection at Mars cannot be treated as an isolated local process, but instead develops within a highly structured induced magnetosphere controlled by IMF draping, crustal magnetic topology, planetary heavy-ion mass loading, and, at low altitudes, ionospheric collisionality.

At the global scale, numerical models have shown that the Martian plasma environment is highly sensitive to the geometry of the external and internal magnetic fields. Simulations of the solar wind interaction with rotating crustal magnetic fields indicate that the large-scale magnetic topology around Mars is not static but varies with the relative configuration between the solar wind and the crustal-field system [31]. This result is relevant to reconnection because such global topological changes can alter where strong magnetic shear and thin current layers are most likely to form [31, 97]. More broadly, multifluid MHD simulations also demonstrate that heavy-ion mass loading from the upper atmosphere and exosphere substantially modifies the large-scale interaction [95]. For example, three-dimensional thermosphere–exosphere inputs are required to reproduce key features of the Martian ionosphere, and hot exospheric oxygen can suppress molecular-ion escape through the mass loading produced by high-altitude O^+ ions [95]. These results imply that reconnection at Mars occurs in a strongly composition-dependent background plasma, and that the efficiency of reconnection-driven energization and escape should depend on both upstream forcing and planetary-ion supply.

The most direct numerical step toward reconnection physics at Mars is provided by a global Hall-MHD model with an embedded PIC treatment of the near-tail current sheet [94]. Such a model is especially important because it bridges the gap between the global induced-magnetotail structure and kinetic-scale reconnection physics. In this framework, the global draped-tail geometry supplies the reconnecting current sheet, while the embedded kinetic description resolves the ion- and electron-scale structure of the reconnection region [94]. This multiscale approach is qualitatively consistent with the MAVEN observations summarized above, which show Hall-like magnetic perturbations, ion outflows involving both protons and planetary heavy ions, and stronger and more systematically organized acceleration of protons than of heavy planetary ions in reconnecting Martian current sheets [63, 64, 66]. The agreement indicates that the Hall-like signatures observed by MAVEN can be interpreted as local kinetic manifestations embedded in a globally formed induced-magnetotail current sheet, rather than as structures generated independently of the large-scale solar wind–Mars interaction.

In addition, global multifluid models of Martian ionospheric electrodynamics indicate that neutral collisions become increasingly important at lower altitudes, particularly in the dynamo region and near crustal-field cusps and loops [98, 99]. More generally, multifluid simulations of magnetic reconnection in partially ionized plasmas demonstrate that ion–neutral coupling, ionization, and recombination can substantially modify current-sheet evolution and reconnection dynamics [100]. These results provide a physical

framework for interpreting the observational transition from more Alfvénic, Hall-like upper-ionospheric events to weaker or more strongly damped signatures deeper in the Martian ionosphere.

These simulations and observations support a multiscale picture in which global solar-wind forcing, IMF draping, and crustal topology prepare reconnection-prone current sheets, whereas local Hall physics, heavy-ion composition, and collisional effects govern the detailed energy conversion and plasma response. A major challenge for future work is therefore to develop self-consistent models that simultaneously include realistic crustal fields, multi-ion chemistry, ionospheric collisions, kinetic reconnection physics, and time-dependent upstream driving [94, 95].

3 Discussions

3.1 Reconfiguration of Mars' magnetosphere by magnetic reconnection

In contrast to Earth, where reconnection operates within a global dipolar system, reconnection at Mars develops in an induced magnetosphere shaped by solar-wind draping, localized crustal magnetic fields, and a heavy-ion-rich ionosphere. Its structural effects are therefore expressed less through classical dipolar substorm behavior and more through the formation, evolution, and disruption of thin current sheets, flux ropes, reconnection-driven jets, and front-like structures. These recent observations show that Mars' magnetosphere is not simply a steady draped-field configuration, but a highly dynamic system that is repeatedly reorganized by reconnection [101].

A global consequence of reconnection is the production of magnetic flux ropes, which represent coherent bundles of reconnected flux and an important mode of magnetic-flux and plasma transport. In the Martian magnetotail, MGS observations revealed chains of flux ropes moving tailward within the current sheet, suggesting that a single reconnection site can generate multiple detached structures through secondary instability or time-dependent reconnection [102]. Statistical studies further suggest that ionospheric flux ropes may arise through more than one pathway, including reconnection between crustal loops and reconnection between crustal fields and the IMF [103, 104]. Structurally, these observations show that reconnection at Mars does not merely alter field-line connectivity locally, but can create detached or semi-detached magnetic entities that redistribute both flux and planetary plasma across the magnetosphere.

Reconnection-driven jets and front-like structures provide further evidence that reconnection actively reorganizes Mars' induced magnetosphere. Bidirectional ion jets observed within current sheets on both the dayside and nightside indicate local conversion of magnetic energy into directed plasma flow and mark the release of magnetic tension from newly reconnected field lines [74]. Their broad distribution suggests that reconnection-driven flow channels are a common component of Martian plasma transport. The detection of a tailward reconnection front in the Martian magnetotail extended this picture further by showing that sharp front-like magnetic structures can form even in an induced magnetotail without a global intrinsic dipole field [105]. Such fronts indicate that reconnection at Mars can generate organized propagating structures that redistribute flux and plasma at ion scales, much as dipolarization-like fronts do in intrinsic magnetospheres, but within a markedly different global magnetic geometry.

These observations support a view of Mars' magnetosphere as a reconnection-driven, structurally variable system. Reconnection thins and fragments current sheets, generates

flux ropes and ejected magnetic structures, and drives jets and fronts that redistribute magnetic flux and plasma. It links kinetic-scale processes to larger-scale reconfiguration of the induced magnetosphere. These structural effects also help set the conditions for plasma escape and atmospheric loss. Global MAVEN maps of electromagnetic energy transport and conversion place these local reconnection processes within the planetary-scale energy budget: the bow shock acts predominantly as a generator, the induced magnetotail as a load, and the magnetosheath alternates between these roles depending on the draping geometry [106]. This system-level picture provides the macroscopic context in which local reconnection can contribute to plasma heating, directed acceleration, magnetic reconfiguration, and atmospheric escape, although the fraction of the total energy conversion attributable specifically to reconnection has not yet been isolated.

3.2 Contributions of magnetic reconnection on ion escape from Mars

Magnetic reconnection likely affects Martian ion escape in two closely coupled ways: by changing magnetic topology and by energizing planetary plasma. At Mars, ion escape is not only determined solely by local acceleration, but also by whether ions can access magnetic pathways that connect the ionosphere to the magnetosheath, the magnetotail, or interplanetary space. Reconnection is therefore important not only as a energy-conversion process, but also as a transport mechanism that opens, closes, and reorganizes escape channels. Therefore, reconnection-driven escape at Mars should be understood as the combined outcome of magnetic linking, ion heating, and subsequent large-scale plasma transport.

On the dayside, reconnection at the induced magnetopause and in crustal-field regions appears capable of providing direct access between ionospheric plasma and the external solar-wind-driven environment. Dayside induced-magnetosphere reconnection can open the boundary between the induced magnetosphere and the magnetosheath, allowing planetary ions to cross into regions where they may be lost through direct acceleration or subsequent pickup by the solar wind [107, 108]. Similarly, in crustal-field regions, reconnection between crustal magnetic fields and the draped IMF can convert previously closed loops into open field lines and thereby provide pathways for plasma release from near-ionospheric source regions [78]. Reconnection at mini-magnetopause boundaries or in more eruptive crustal-field events may drive localized plasma release on top of this basic opening process. These dayside reconnection are potentially important for ion escape, even if their global contribution remains difficult to quantify from the currently limited event statistics.

On the nightside, magnetotail reconnection provides a complementary escape pathway. In the induced magnetotail, reconnection converts magnetic energy into tailward plasma flow, restructures the cross-tail current sheet, and generates ion jets, flux ropes, and reconnection fronts that transport plasma away from the planet. Statistical studies suggest that reconnecting current sheets are associated with enhanced tailward proton fluxes, indicating that magnetotail reconnection likely contributes directly to proton escape. The role of tail reconnection in heavy-ion loss appears more complex. Statistical studies indicate that heavy ions can be heated and transported in reconnecting structures, but their statistical response is less systematic than that of protons [67, 69]. This likely reflects the multicomponent nature of the Martian plasma sheet, where heavy ions are only weakly magnetized relative to proton-scale current sheets and may therefore

couple less efficiently to reconnection electric fields. Magnetotail reconnection is thus best viewed as an important escape channel whose effectiveness depends on both magnetic structure and plasma composition.

The ionosphere itself may also contribute directly to escape through reconnection, particularly in the upper ionosphere where collisionless or weakly collisional reconnection can accelerate cold planetary ions into field-aligned outflows [90]. In this setting, ionospheric reconnection may function less as an isolated escape mechanism and more as a pre-acceleration stage that feeds ions into larger-scale escape pathways, including transport into the magnetotail or pickup by the solar wind. This is especially relevant for heavy ions from Mars, which dominate much of Mars' ionosphere but require substantial energization before they can escape. By contrast, at lower altitudes increasing ion-neutral collisionality is expected to reduce outflow speeds and suppress direct reconnection-driven escape. The efficiency of ionospheric reconnection as an escape process therefore depends strongly on altitude and collisional regime.

Overall, recent observations suggest that reconnection contributes to Martian ion escape through multiple pathways rather than through a single dominant channel. Its effects include opening the dayside boundary, releasing plasma from crustal-field regions, driving tailward outflow in the magnetotail, and pre-accelerating thermal ions in the upper ionosphere. Reconnection is also likely to become more effective during periods of strong IMF variability or disturbed solar-wind forcing, when favorable magnetic shear and rapid magnetospheric reconfiguration occur more often. For early Mars, when the young Sun likely drove a denser and more variable solar wind, such reconnection-driven escape may have been substantially enhanced. However, the cumulative contribution of reconnection to the long-term Martian ion loss budget remains uncertain, particularly in comparison with other escape channels such as pickup, sputtering, and ambipolar outflow. A central task for future work is therefore to determine when reconnection merely redistributes plasma within the Martian system and when it produces net loss to interplanetary space.

3.3 Heavy-ion effects on magnetic reconnection

A defining property of magnetic reconnection at Mars is that it often develops in a plasma environment rich in planetary heavy ions rather than in a predominantly proton plasma. Mars' ionosphere is dominated by heavy ions such as O^+, O_2^+, and CO_2^+, which also populate the induced magnetosphere, crustal-field regions, and magnetotail current sheet [10, 109]. Reconnection at Mars differs in an important respect from the more proton-dominated reconnection environments typically discussed for Earth's magnetosphere. In this sense, Mars provides a natural laboratory for examining how heavy ions modify collisionless reconnection.

The most basic effect of heavy ions is mass loading. By increasing the plasma mass density, heavy ions reduce the characteristic Alfvén speed and therefore modify reconnection outflow speeds, current-sheet scales, and energy conversion rates [110]. In a multi-ion plasma, protons and heavy ions are not expected to demagnetize in the same way or on the same scales. This introduces a more complex kinetic hierarchy than in the standard electron–proton picture, in which the ion diffusion region is usually characterized by a single dominant ion scale. In principle, a heavy-ion-rich plasma can support broader ion-scale structure, altered Hall fields, and species-dependent exhaust

dynamics. Although the full multiscale diffusion-region hierarchy predicted by generalized multi-fluid or kinetic theory has not yet been resolved directly at Mars, previous observations are broadly consistent with the idea that heavy ions modify both the structure and the efficiency of reconnection [67, 69].

The clearest observational evidence for this effect comes from the induced magnetotail. MAVEN observations show that proton outflows in reconnecting current sheets are generally stronger and more clearly organized than those of heavy ions, whereas the response of O^+ and O_2^+ is weaker and more variable [66, 67, 69]. This behavior is consistent with the expectation that lighter ions remain more effectively coupled to proton-scale reconnection structures, while heavier ions, with larger inertial lengths and gyroradii, are less efficiently accelerated by the same electric fields. Similar asymmetry may also affect boundary reconnection and upper-ionospheric reconnection, where the plasma on one side of the current sheet is often strongly enriched in planetary heavy ions. In such cases, reconnection should be regarded not simply as asymmetric in density, but also as asymmetric in ion composition.

4 Knowns and unknowns

Recent studies, based on measurements from MGS, Mars Express, MAVEN, and Tianwen-1, have shown that reconnection can occur in multiple regions, including the dayside induced magnetosphere [61, 78], the induced magnetotail [66], crustal-field regions [78–81], and the upper ionosphere [78, 82]. Across these regions, reconnection can restructure current sheets, generate flux ropes and reconnection fronts, accelerate plasma, open magnetic escape pathways, and influence auroral activity and ion loss. Reconnection at Mars differs from more familiar terrestrial cases because of its induced global topology, strong crustal-field asymmetry, partial ionospheric collisionality, and substantial planetary heavy-ion component.

Despite these advances, several key questions still remain unresolved.

The occurrence rate of reconnection in different Martian regions is still not well quantified. Magnetotail reconnection has been studied statistically, but dayside induced-magnetosphere reconnection, crustal-field reconnection, and ionospheric reconnection are still largely constrained by limited events. Future statistical surveys should determine how reconnection occurrence depends on solar zenith angle, altitude, crustal-field strength, local time, solar cycle, IMF orientation, and upstream solar-wind conditions.

The contribution of reconnection to the global ion escape budget remains uncertain. Reconnection can accelerate ions locally and open magnetic pathways to the magnetosheath, magnetotail, or solar wind, but it is not always clear whether local energization leads to net atmospheric loss. A central future task is to distinguish between plasma redistribution within the Martian system and true escape to interplanetary space, especially for heavy ions.

The role of heavy ions in modifying reconnection physics requires further study. Planetary heavy ions can alter Alfvén speeds, current-sheet scales, Hall signatures, outflow properties, and energy partition. Existing observations suggest species-dependent acceleration, particularly in the magnetotail, but it remains unclear how heavy ions modify the reconnection rate, diffusion-region structure, and coupling between kinetic-scale processes and global plasma transport. Realistic multi-fluid, hybrid, and kinetic simulations with Martian ion composition are needed to address this problem.

The transition between collisionless and collisional reconnection in the Martian ionosphere remains poorly constrained. At higher altitudes, reconnection can display Hall-like signatures and jet-like outflows, whereas at lower altitudes ion-neutral collisions may suppress or strongly modify reconnection. Future work should clarify whether reconnection in the lower ionosphere can still contribute to heating, vertical plasma transport, or magnetic topology change even when fast outflows are absent.

The coupling between IMF variability and local magnetic topology is still not fully understood. Recent multipoint observations suggest that IMF rotations can trigger reconnection in both the magnetotail and the dayside upper ionosphere. However, the relative roles of upstream forcing, internal current-sheet evolution, crustal-field geometry, and ionospheric plasma conditions remain difficult to separate. Coordinated upstream and near-Mars measurements will be essential for determining how rapidly the Martian plasma environment responds to IMF changes.

The global spatial scale and system-level influence of Martian reconnection remain largely unknown. Most current observations identify reconnection locally, but they do not determine how large the reconnection region is, how far reconnection-generated structures propagate, or how much of the induced magnetosphere is affected by a given event. Future multi-spacecraft observations are therefore crucial [101]. Simultaneous measurements at different locations around Mars could reveal the spatial extent of reconnection X-lines, the propagation range of jets, flux ropes, and reconnection fronts, and the degree to which local reconnection events reorganize the global induced magnetosphere. Such joint observations would help bridge the gap between local kinetic signatures and planetary-scale plasma transport.

Addressing these questions will require a combination of long-term statistical surveys, coordinated multi-spacecraft observations, and numerical modeling. Global MHD and hybrid simulations are needed to capture IMF draping, crustal-field topology, and magnetotail reconfiguration, while kinetic simulations are required to resolve diffusion-region physics, heavy-ion effects, and collisional processes. Upcoming missions such as NASA's ESCAPADE [111] and JAXA's Mars Moons eXploration (MMX) [112], together with Tianwen-1, Mars Express, and possible next-generation orbiters, will be central to building a more complete picture of reconnection at Mars.

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Author contributions
R. L. wrote the main manuscript text. Y. W., S. H., K. J., Q. X. reviewed the manuscript.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable. This is a theoretical study, and the study did not involve human or animal subjects, their data, or biological materials. This study has been conducted under the highest ethical standards.

Consent for publication
Not applicable. This is a theoretical study, and the study did not involve human or animal subjects, their data, or biological materials. This study has been conducted under the highest ethical standards.

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The authors declare no competing interests.

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